

1	(a) (gravitational) force proportional to product of masses and inversely proportional to square of separation reference to <i>either</i> point masses <i>or</i> particles <i>or</i> 'size' much less than separation	M1	A1	[2]
	(b) gravitational force provides/is the centripetal force $GM_N m / r^2 = m r \omega^2$ (or $m v^2 / r$) $2\pi / T$ (or $v = 2\pi r / T$) leading to $GM_N = 4\pi^2 r^3 / T^2$	B1	M1	A1
	(c) $M_N / M_U = (3.55 / 5.83)^3 \times (13.5 / 5.9)^2$ x^3 factor correct T^2 factor correct ratio = 1.18 (<i>allow 1.2</i>)	C1	C1	A1
	<i>alternative method:</i> mass of Neptune = 1.019×10^{26} kg mass of Uranus = 8.621×10^{25} kg ratio = 1.18	(C1)	(C1)	(A1) [3]